

Aayush Raj Pandey

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Education

The College of Idaho — Caldwell, Idaho

Bachelor's Degree in Physics and Mathematics

Expected Graduation: 2027

- Relevant Coursework: Statistical Machine Learning, Data Analysis, Physics Modeling, Linear Algebra, Differential Equations
 - Activities: Student Senator, Soccer Club President, ISO Executive Council, Student Research Conference Participant
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Technical Skills

Programming & Data Science: Python, pandas, NumPy, scikit-learn, matplotlib, Jupyter Notebook, Quarto

Machine Learning & Analytics: Logistic Regression, Random Forest, XGBoost, Feature Engineering, Cross-Validation, ROC-AUC Evaluation

Tools & Platforms: GitHub, VSCode, Positron, Excel, LaTeX

Business & Leadership: Project Management, Event Coordination, Strategic Planning, Stakeholder Communication

Projects

NBA Win Prediction Modeling | Python, scikit-learn, pandas

- Built a machine learning pipeline to predict NBA game outcomes using modern NBA data from 2021–2026.
- Engineered rolling matchup differential features including recent win percentage, offensive averages, defensive averages, and point differential trends.
- Compared Logistic Regression, Ridge, Lasso, Random Forest, and XGBoost models using temporal train-test validation.
- Achieved strongest predictive performance using tuned Logistic Regression with ROC-AUC evaluation.
- Demonstrated that feature engineering contributed more predictive value than increased model complexity.

MH370 Search Physics Modeling Project | Python, Physics Modeling

- Developed a physics-based capstone project modeling potential MH370 crash and debris search regions.
 - Incorporated Bayesian search theory, aircraft motion equations, ocean drift modeling, and probability-based search optimization.
 - Applied machine learning and mathematical modeling concepts to real-world aviation search analysis.
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Professional Experience

HR Assistant

The College of Idaho — Caldwell, Idaho
February 2024 – Present

- Maintain employee records and assist with documentation processing and policy compliance.
- Support filing systems and administrative operations for human resources workflows.
- Collaborate with staff to improve organization and operational efficiency.

Events Associate

The College of Idaho — Caldwell, Idaho
August 2024 – Present

- Coordinate logistics, scheduling, lighting, and sound operations for campus-wide events.
- Assist with vendor coordination and event execution for large student gatherings.
- Support event planning initiatives that improve campus engagement and participation.

Founder & Partner

Evoliption — Remote/New York
2025 – Present

- Co-founded a cosmetics business collaborating with more than 100 medical spas across New York.
- Led partnership negotiations, business development, and client acquisition strategies.
- Generated approximately \$10,000 in profit within the first few months of operation.
- Oversaw operations, customer relations, and strategic expansion planning.

Project Manager

Adventures Magic Trek — Kathmandu, Nepal
January 2021 – March 2023

- Managed tourism and trekking projects while overseeing logistics, budgeting, and stakeholder communication.

- Coordinated vendor negotiations and customer relations to improve operational effectiveness.
 - Implemented marketing initiatives that increased customer engagement and visibility.
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Leadership & Activities

Student Senator

The College of Idaho Student Senate

2023 – Present

- Elected representative responsible for campus-wide budgeting and student policy decisions.
- Evaluate funding proposals and collaborate with faculty and administration on institutional initiatives.
- Represent student interests in governance and strategic campus discussions.

President, Soccer Club

The College of Idaho

August 2023 – Present

- Lead and manage the college Soccer Club, overseeing operations, budgeting, recruitment, and event coordination.
- Organize practices, scrimmages, and intercollegiate competitions.
- Foster team culture, leadership development, and student involvement.

Annotator

Sigma Chi — The College of Idaho

December 2024 – Present

- Maintained organizational documentation and financial records.
- Assisted leadership with budgeting, alumni relations, and event planning.
- Helped support organizational financial recovery and long-term alumni engagement.

Founder

Discite Educational Initiative

2023 – Present

- Founded a nonprofit initiative supporting students from rural communities interested in STEM education.
 - Connect students with professors, mentors, and research opportunities.
 - Designed educational outreach programs focused on physics and academic development.
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Additional Experience

Volunteer Coordinator

Rotaract Club of Rajdhani — Kathmandu, Nepal

- Organized fundraising and community outreach initiatives.
- Managed volunteer recruitment and event coordination.

CANSAT Program Participant

Kathmandu, Nepal

2017 – 2018

- Participated in aerospace and satellite engineering activities.
 - One of the first students in Nepal to communicate directly with an astronaut aboard the International Space Station.
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Languages

- English (Fluent)
- Nepali (Fluent)
- Spanish (Intermediate)